Guided Tutorial – Agents and PEAS

PEAS

The PEAS framework helps describe the components of an agent's task environment: Performance measure, Environment, Actuators, and Sensors.

Activity:

1. Check following YouTube videos of intelligent drone products and give a PEAS description for an intelligent delivery drone and identify characteristics of its task environment.

   DJI FlyCart 30: https://www.youtube.com/watch?v=Hhp11I-vGHA
   Skydio R1 Autonomous Drone https://www.youtube.com/watch?v=i1ILgLkWaLI

Possible Solution:

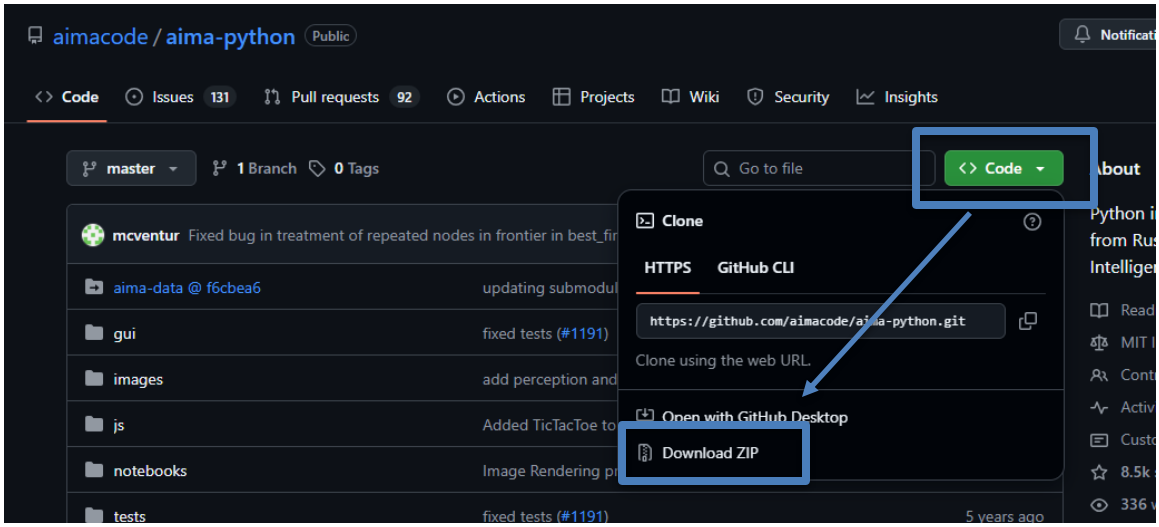
• Performance measure: On-time delivery, safe navigation, energy efficiency, minimal collisions.
• Environment: Urban areas, weather conditions, air traffic, delivery addresses.
• Actuators: Rotors (for movement), robotic arm (for parcel drop), communication system.
• Sensors: GPS, cameras, obstacle detection sensors, battery monitor.

Agents

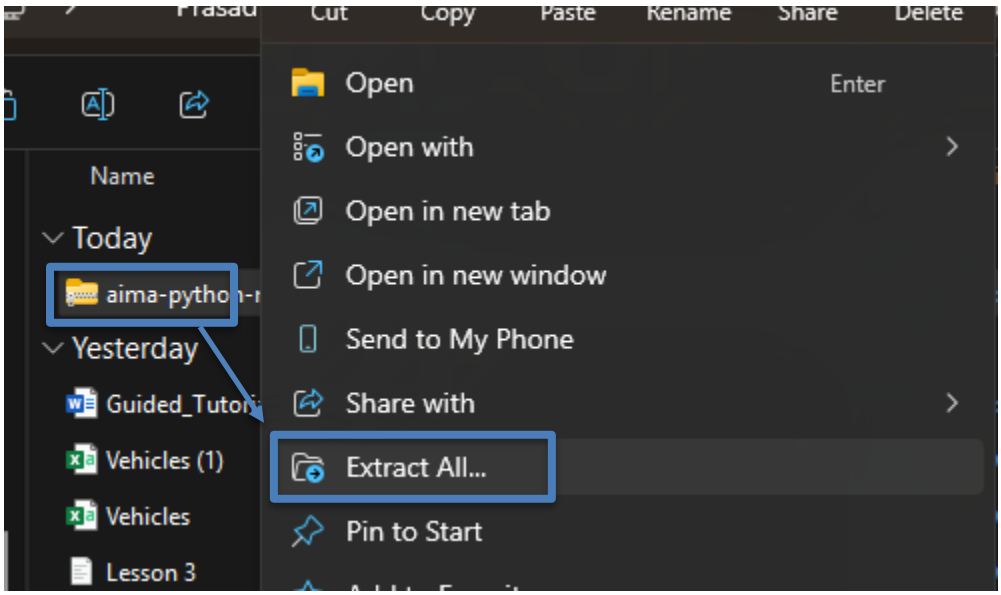
An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment through actuators.

Activity: To be familiar with Anaconda Jupyter lab

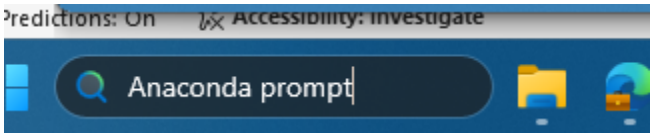
1. Go to https://github.com/aimacode/aima-python
2. Download the Git folder

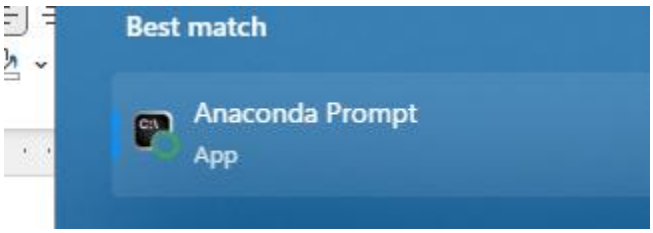


3. Extract the Zip file



4. Open Anaconda Prompt





This will open a terminal but if you do not have anaconda navigator then install it.

Note: To learn how to install anaconda navigator click on this link https://www.youtube.com/watch?v=11Qwh-pkNNc

5. Type **conda create -n py37 python=3.7**
   And type **y** and enter to proceed

```
The following NEW packages will be INSTALLED:

  ca-certificates    pkgs/main/win-64::ca-certificates-2025.9.9-haa95532_0
  certifi            pkgs/main/win-64::certifi-2022.12.7-py37haa95532_0
  openssl            pkgs/main/win-64::openssl-1.1.1w-h2bbff1b_0
  pip                pkgs/main/win-64::pip-22.3.1-py37haa95532_0
  python             pkgs/main/win-64::python-3.7.16-h6244533_0
  setuptools         pkgs/main/win-64::setuptools-65.6.3-py37haa95532_0
  sqlite             pkgs/main/win-64::sqlite-3.50.2-hda9a48d_1
  ucrt               pkgs/main/win-64::ucrt-10.0.22621.0-haa95532_0
  vc                 pkgs/main/win-64::vc-14.3-h2df5915_10
  vc14_runtime       pkgs/main/win-64::vc14_runtime-14.44.35208-h4927774_10
  vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.44.35208-ha6b5a95_10
  wheel              pkgs/main/win-64::wheel-0.38.4-py37haa95532_0
  wincertstore       pkgs/main/win-64::wincertstore-0.2-py37haa95532_2

Proceed ([y]/n)? y

Downloading and Extracting Packages:
```

6. Type **conda activate py37**
7. Type **conda install jupyterlab** and **y** to proceed
8. Change the current working directory in Anaconda Prompt to your downloaded extracted folder above
   Copy the path of the extracted folder
   Example: C:\Users\Prasad Pushpika\Downloads\aima-python-master\aima-python-master
   Then type the path after cd like below example
   cd C:\Users\Prasad Pushpika\Downloads\aima-python-master\aima-python-master
9. Then type **pip install -r requirements.txt**

Note: It will install few libraries and there could be some errors occurs due to version updates **but move forward to next step**

```
(py37) C:\Users\Prasad Pushpika\Downloads\aima-python-master\aima-python-master>pip install -r requirements.txt
Collecting cvxopt
  Downloading cvxopt-1.3.2.tar.gz (4.1 MB)
     ---------------------------------------- 4.1/4.1 MB 1.0 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
  Preparing metadata (pyproject.toml) ... done
Collecting image
  Downloading image-1.5.33.tar.gz (15 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: ipython in c:\users\prasad pushpika\.conda\envs\py37\lib\site-packages (from -r requirements.txt (line 3)) (7.31.1)
Collecting ipythonblocks
  Downloading ipythonblocks-1.9.1-py2.py3-none-any.whl (14 kB)
Collecting ipywidgets
  Downloading ipywidgets-8.1.7-py3-none-any.whl (139 kB)
     ---------------------------------------- 139.8/139.8 kB 2.1 MB/s eta 0:00:00
Collecting jupyter
```

10. Then type **jupyter lab**

```
Read the migration plan to Notebook 7 to learn about the new features and the actions to take if you are using extension
s.

https://jupyter-notebook.readthedocs.io/en/latest/migrate_to_notebook7.html

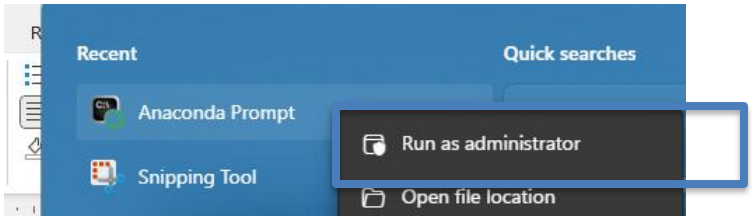
Please note that updating to Notebook 7 might break some of your extensions.

[I 2025-09-27 16:13:14.973 ServerApp] nbclassic | extension was successfully loaded.
[I 2025-09-27 16:13:14.973 ServerApp] Serving notebooks from local directory: C:\Users\Prasad Pushpika\Downloads\aima-py
thon-master\aima-python-master
[I 2025-09-27 16:13:14.973 ServerApp] Jupyter Server 1.23.4 is running at:
[I 2025-09-27 16:13:14.973 ServerApp] http://localhost:8888/lab?token=570baca61d344208dd04a05ebc93033c5c50056569e2d2fa
[I 2025-09-27 16:13:14.973 ServerApp]  or http://127.0.0.1:8888/lab?token=570baca61d344208dd04a05ebc93033c5c50056569e2d2
fa
[I 2025-09-27 16:13:14.973 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirm
ation).
[E 2025-09-27 16:13:14.973 ServerApp] Failed to write server-info to C:\Users\Prasad Pushpika\AppData\Roaming\jupyter\ru
ntime\jpserver-6968.json: [Errno 13] Permission denied: 'C:\\Users\\Prasad Pushpika\\AppData\\Roaming\\jupyter\\runtime\
\jpserver-6968.json'
Traceback (most recent call last):
  File "C:\Users\Prasad Pushpika\.conda\envs\py37\Scripts\jupyter-lab-script.py", line 10, in <module>
```

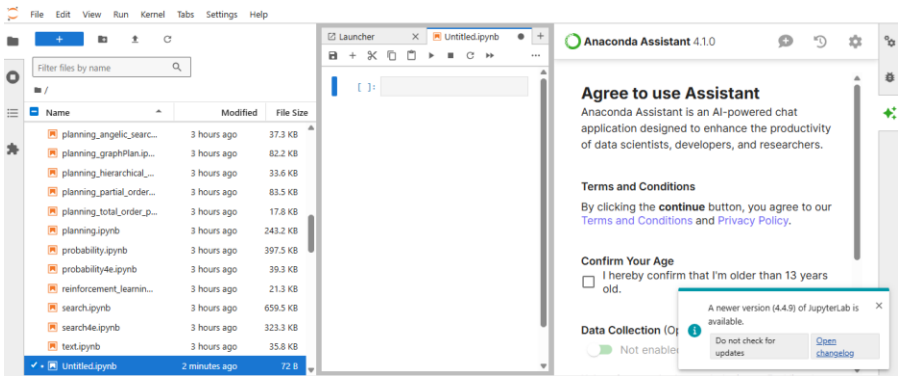
11. Then open on the **localhost link** in web browser that shows in the terminal

    Note: If it gives an error message Please run anaconda prompt using **Run as Administrator** and then again **change the current working directory** to extracted folder (example: **cd C:\Users\Prasad Pushpika\Downloads\aima-python-master\aima-python-master)**

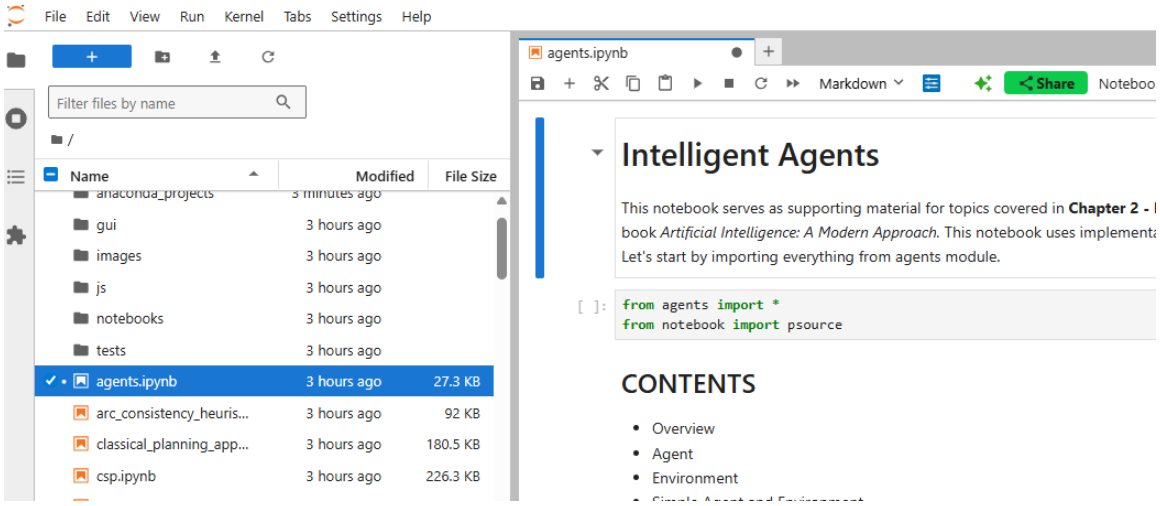
    Then type **jupyter lab** and copy the localhost link and use web browser to open the jupyter lab



12. Then you can open the jupyter lab

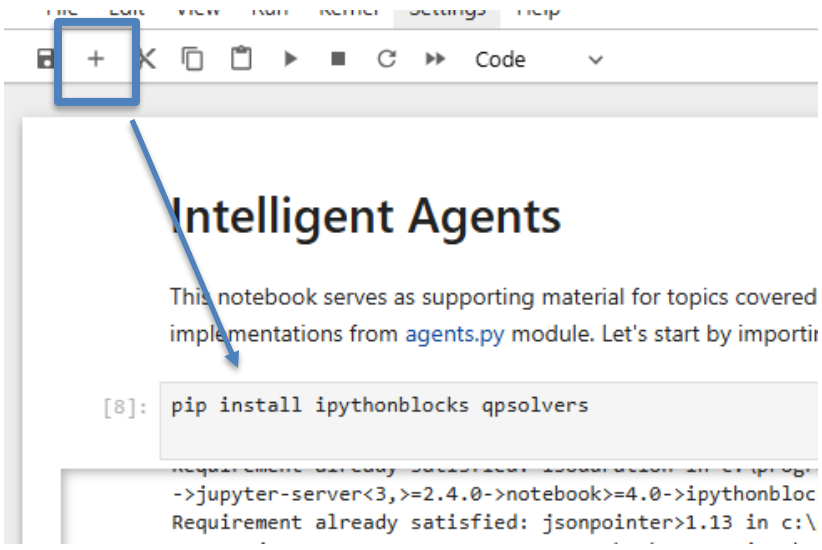


13. Now open **agents.ipynb**



14. **Now try to run the code and understand (self-study) how you are going to implement and agent and its environment using these codes.**

Note: if you are getting errors in missing python libraries then try to install them by opening a new code cell.
Example is given below which shows how to install **ipythonblocks** and **qpsolvers** libraries



Note: If it says missing CSV files in the **aima-data** folder. Then download them from this github link: https://github.com/aimacode/aima-data/tree/f6cbea61ad0c21c6b7be826d17af5a8d3a7c2c86
And move them to the folder called **aima-data** in the previously extracted folder and try to execute again.